

AutoPlanner 高级天然产物全流程专家汇报

01

以 bufotalin stitched semisynthesis 为完整运行示例

Target

name: bufotalin

formula: C₂₆H₃₆O₆

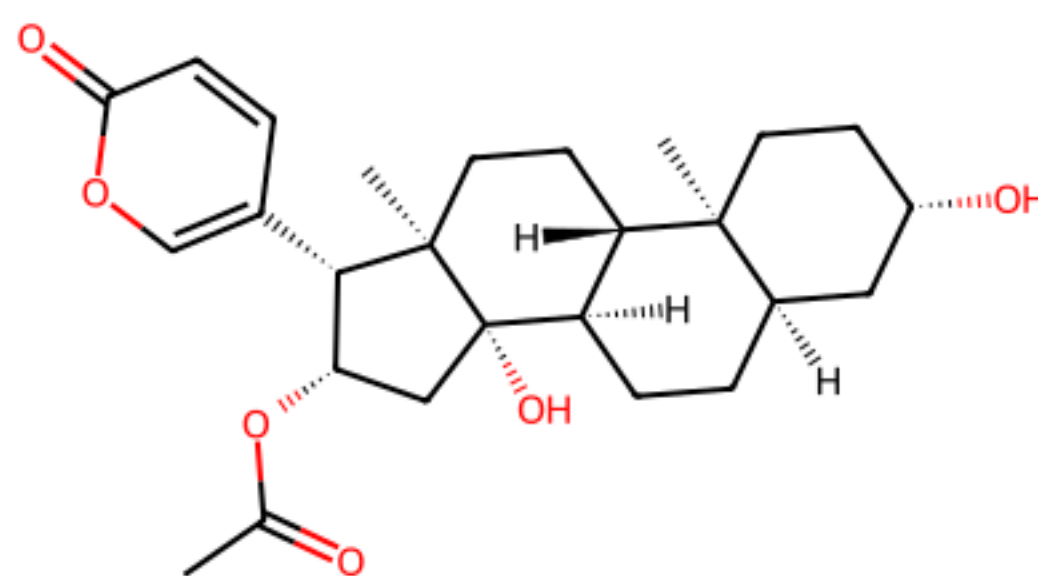
exact MW: 444.251

heavy atoms / rings: 32 / 5

stereocenters: 9

InChIKey: VOZHMAYHYHEWBW-NVOOAVKYSA-N

risk flags: polycyclic_or_steroid_like



Bufotalin RDKit depiction

Planner

accepted

canonical_agent_controller

Source chain

15 steps

terminal compound 11

Subgoal

20/56 accepted

best rank 2

Final

solved

solved=True

Executive conclusion

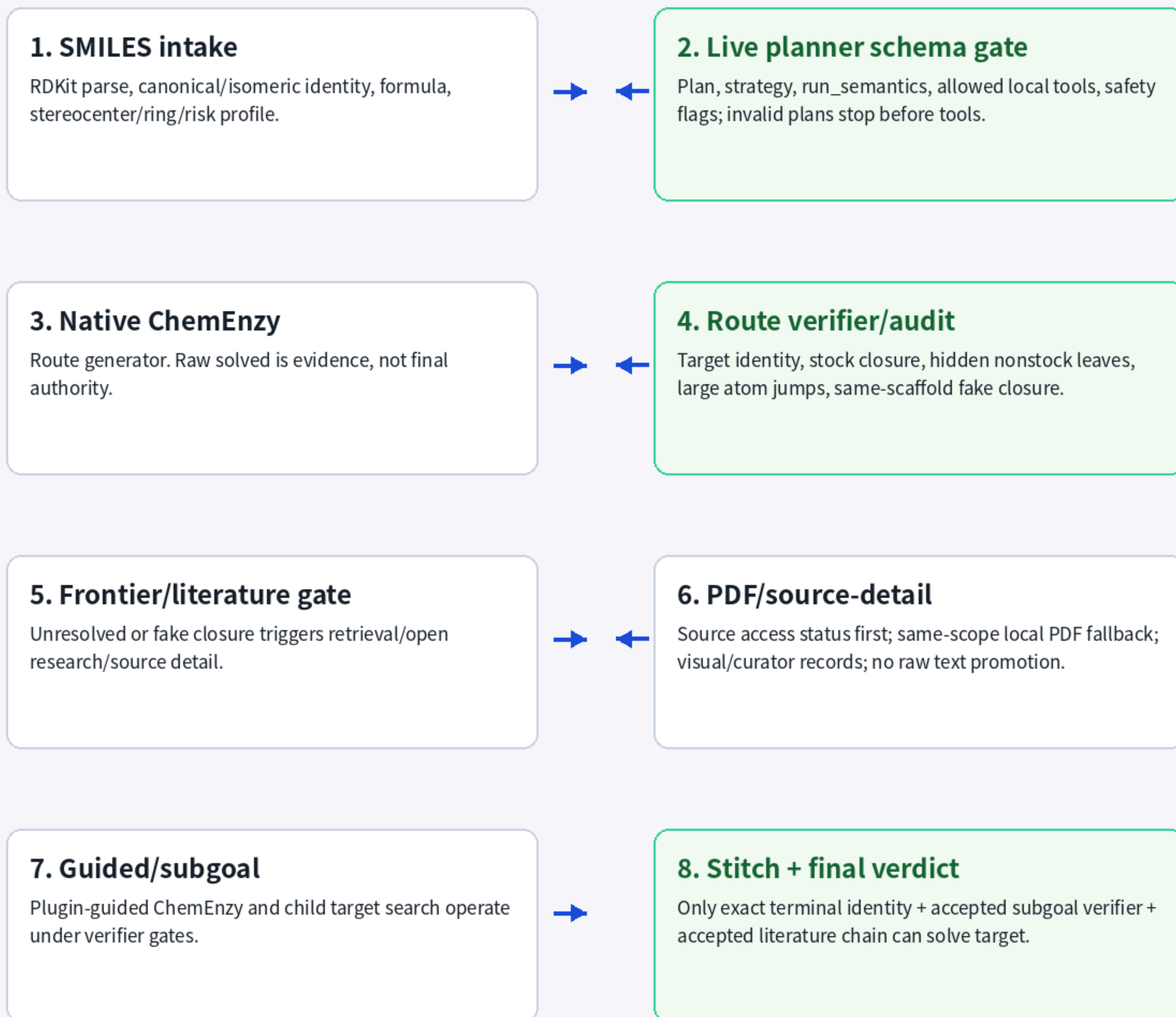
本轮 live planner 通过 schema gate, 未使用 fallback plan 作为判断基础。

PDF 使用的是本地既有文件, 但 PDF render、crop、visual audit、source-detail validation、subgoal search 和 stitch audit 均重新执行。

最终 solved 不是 ChemEnzy 原生/guided 的直接输出, 而是 15 步 source-detail 文献链与 compound 11 子目标 solved route 的 deterministic stitch。

Combined route: 24 steps = 9 subgoal + 15 literature.

Typed tools + deterministic gates



Architecture invariant

Planner 可以发起工具和子目标搜索，但 solved 权威只来自 deterministic validators。

子目标 solved 只说明该 terminal 可解；必须由 stitch audit 证明 terminal 与文献链末端完全同一分子，才能上升为目标 solved。

Why live planner matters

Live planner result

accepted: True
elapsed_s: 42.189
raw_run_semantics_type: str
run_semantics: canonical_agent_controller
run_semantics_changed: False
planned tool count: 11
record:
/root/autodl-tmp/AutoPlanner/results/shared/bufotalin_stitched_fullflow_existing_pdf_20260608_0345/codex_planner_run_record.json

Fixed contract

run_semantics 现在支持安全 normalization audit, 但 prompt 明确要求输出字符串枚举。
allowed_local_tools 新增
stitch_literature_chain_with_subgoal_route。
deterministic fallback 不再作为本次判断路径；完整示例来自 live planner accepted run。

Planned tools from live planner

1. run_smiles_first_literature_workflow
2. run_open_structure_research_agent
3. extract_pdf_literature_structures
4. validate_literature_intermediate_chain
5. build_source_detail_curator_records
6. compile_source_detail_chain_route
7. audit_route_and_extract_frontier
8. run_route_expansion_subgoal_search
9. stitch_literature_chain_with_subgoal_route
10. validate_artifact_bundle
11. emit_final_verdict

Subgoal/stitch responsibility

Agent 发起子目标搜索；最终审计和拼接也作为工具链的一部分执行。
stitch 工具要求 target identity audit、terminal/subgoal exact match、subgoal route verifier accepted、source-detail chain accepted。
任何单独的 subgoal solved 都不会自动写成 target solved。

Native ChemEnzy

elapsed_s: 303.173
route_count: 179
accepted_route_count: 0
route_status: fake_closed_rejected
reasons: advanced_same_scaffold_terminal,
large_atom_jump,
no_verifier_accepted_stock_closed_route

Open research

elapsed_s: 930.123
status: timeout
reasons: open_structure_research_timeout
output_dir:
/root/autodl-tmp/AutoPlanner/results/shared/bufotalin_stitched_fullflow_existing_pdf_20260608_0345/open_structure_research

Patched guided rerun

elapsed_s: 327.683
policy_id: bufotalin_literature_plugin_only
rerun_reason:
compiled_literature_template_plugin_available
route_status: fake_closed_rejected
solved: False
verifier reasons: advanced_same_scaffold_terminal,
large_atom_jump,
no_verifier_accepted_stock_closed_route

Route expansion subgoal

target: compound 11 exact TET2025 terminal
route_status: solved
route_count: 56
accepted_route_count: 20
best_route_rank: 2
target_match: True

Tool-call sequence

1. run_chemenzy | accepted | 303.173s | -
2. audit_route_and_extract_frontier | accepted | 1.676s | -
3. run_smiles_first_literature_workflow | accepted | 0.377s | -
4. run_open_structure_research_agent | rejected | 930.123s | open_structure_research_timeout
5. extract_pdf_literature_structures | accepted | 0.95s | -
6. validate_literature_intermediate_chain | accepted | 0.056s | -
7. build_source_detail_curator_records | accepted | 0.121s | -
8. compile_source_detail_chain_route | accepted | 0.398s | -
9. run_guided_chemenzy_rerun | accepted | 0s | guided_policy_missing
10. run_route_expansion_subgoal_search | accepted | 501.542s | -
11. stitch_literature_chain_with_subgoal_route | accepted | 0.017s | -

...

AutoPlanner harness expert report

15-step chain from bufotalin to compound 11

Chain validation

accepted: True

step_count: 15

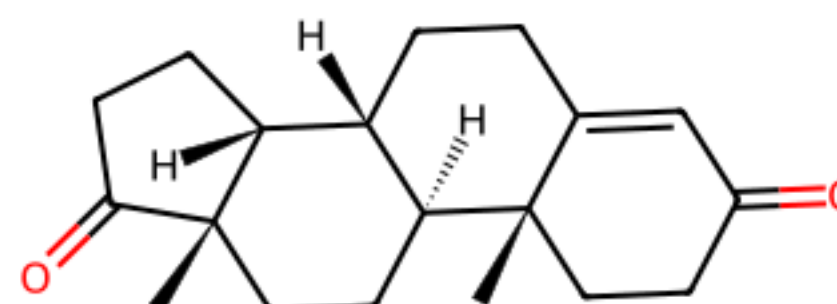
terminal_reached: True

terminal_name: compound 11 exact tet2025 terminal

terminal_smiles:

```
C[C@]12CCC(=O)C=C1CC[C@H]1[C@H]3CCC(=O)[C@@]3(C)CC[C@@H]12
```

reasons: -



Compound 11 terminal

First source-detail steps

1. tet2025_33_to_bufotalin_deprotection
2. tet2025_32_to_33_acetylation
3. tet2025_31_to_32_nabh4_reduction
4. tet2025_30_to_31_tmsof_rearrangement

Last source-detail steps

12. tet2025_23_to_26_c16_bromination
13. tet2025_25_to_23_kselectride_reduction
14. tet2025_24_to_25_hydrogenation
15. tet2025_11_to_24_c17_ketalization

Why this is not yet final solved by itself

The literature chain runs backward from bufotalin to terminal compound 11 and proves a source-detail terminal.

It still requires an independent route closure from stock/building blocks to compound 11.

The stitch tool is the authority that joins these two segments after exact identity audit.

Subgoal search 与 stitched solved

07

Where solved is earned

Subgoal verifier

accepted: True
route_status: solved
route_count: 56
accepted_route_count: 20
best_route_rank: 2
reasons/warnings: large_atom_jump

Terminal identity audit

accepted: True
match_basis: canonical_isomeric_smiles_and_inchikey
terminal InChIKey: AEMFNILZOJDQLW-MGGVOPSTSA-N
subgoal InChIKey: AEMFNILZOJDQLW-MGGVOPSTSA-N
reasons: -

Combined stitched route

Subgoal stock closure: 9 steps



Literature chain: 15 steps

Direction: stock_to_subgoal_terminal_then_literature_to_target | total: 24 steps

Stitched route verdict

accepted: True
solved: True
route_status: solved
stock_audit_passed: True
combined_step_count: 24
warnings: subgoal_verifier:large_atom_jump

Source policy

terminal_identity_match_required: True
subgoal_segment_requires_route_verifier: True
literature_segment_requires_source_detail_chain: True
final_verdict_authority: deterministic_validators

Why the final solved claim is not overbroad

Native ChemEnzy rejected

route_status: fake_closed_rejected
accepted_route_count: 0
reasons: advanced_same_scaffold_terminal,
large_atom_jump,
no_verifier_accepted_stock_closed_route
Interpretation: route generator output contained fake
closure / advanced same-scaffold terminal evidence.

Patched guided rerun rejected

elapsed_s: 327.683
route_status: fake_closed_rejected
accepted_route_count: 0
reasons: advanced_same_scaffold_terminal,
large_atom_jump,
no_verifier_accepted_stock_closed_route
Interpretation: plugin guidance ran, but did not by itself
solve the target.

Artifact bundle validation

accepted: False
reasons: fake_closure_evidence_present,
open_structure_research_timeout
This preserves negative evidence instead of hiding it behind
the stitched solved route.

Final verdict precedence

verdict: solved
solved: True
route_status: solved
The final verdict can be solved because
stitched_semisynthesis_route passed exact identity and
stock audits.

Expert interpretation

Native/guided route failures remain visible and auditable.

The positive claim is narrower: a semisynthesis route made from a verified stock-to-compound-11 subroute and an accepted source-detail literature chain from compound 11 to bufotalin.

Subgoal route warnings such as large_atom_jump are carried as warnings because the verifier still found accepted stock-closed routes.

Every claim has a local artifact path

Key artifact paths

summary: run_dir/bufotalin_stitched_fullflow_summary.json

planner_record: run_dir/codex_planner_run_record.json

pdf_evidence: run_dir/literature_pdf_structure_extraction/literature_pdf_structure_evidence.json

visual_chain_validation: run_dir/literature_intermediate_chain_validation/visual_structure_chain_validation.json

source_detail_chain_route: run_dir/source_detail_chain_route/source_detail_route_chain_audit.json

route_expansion_subgoals: run_dir/route_expansion_subgoals

stitched_route: run_dir/stitched_semisynthesis_route/stitched_semisynthesis_route.json

final_verdict: run_dir/final_verdict.json

patched_guided_tool_call: run_dir/guided_chemenzy_patched_tool_call.json

guided_route_verifier: run_dir/guided_route_verifier_report.json

artifact_bundle_validation: run_dir/artifact_bundle_validation.json

Code paths changed

schemas.py: run_semantics normalization + stitch tool registration

codex_plan.py: planner prompt/schema and stitch planning instruction

tools.py: stitch handler, plugin-only guided policy, compiled_downstream state publication

runner.py: final verdict can consume stitched solved route

literature_pdf_extraction.py: robust crop source fallback

Verification

pytest selected harness contract tests passed.

Full bufotalin run completed with live planner accepted.

Patched guided supplement completed as a real ChemEnzy run and was rejected by verifier.

Report scope

This PDF is an expert briefing, not a laboratory procedure. It reports architecture, typed evidence flow, and deterministic audit outcomes.